QUASIHARMONIC MODELING OF THERMAL EXPANSION IN MATERIALS

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Introduction

- **Temperature** drives thermal expansion, which alters conductivity in thermoelectric materials and electronic coupling in organic semiconductors.
- **High-performance computing (HPC)** with **Density Functional Theory (DFT)** in CRYSTAL 23 is used to model these effects and reveal how thermal expansion influences electronic structure and charge transport.

Theory

DFT: PBE

$$zT = \frac{\sigma S^2 T}{\kappa}$$

$$\kappa = \kappa_e + \kappa_p$$

$$[\sigma]_{qr}(\mu, T) = e^2 \int dE \left(-\frac{\partial f_0}{\partial E}\right) \Xi_{qr}(E)$$

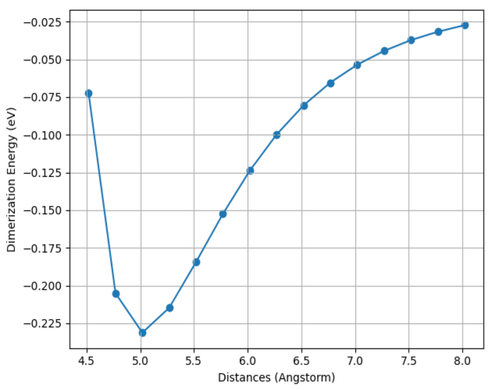
$$[\sigma S]_{qr}(\mu, T) = \frac{e}{T} \int dE \left(-\frac{\partial f_0}{\partial E}\right) (E - \mu) \Xi_{qr}(E)$$

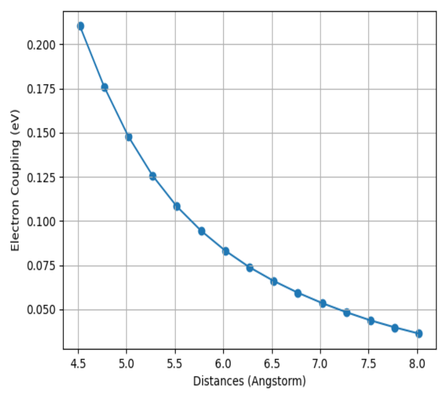
$$[\kappa_e]_{qr}(\mu, T) = \frac{1}{T} \int dE \left(-\frac{\partial f_0}{\partial E}\right) (E - \mu)^2 \Xi_{qr}(E)$$

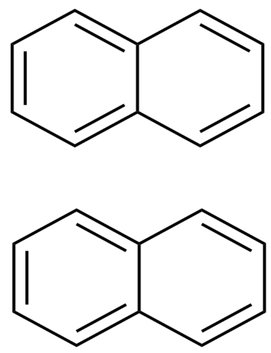
$$\Xi_{qr}(E) = \tau \sum_{\mathbf{k}} \frac{1}{N_{\mathbf{k}}} \frac{1}{V} \sum_{i,j} v_{i,q}(\mathbf{k}) v_{j,r}(\mathbf{k}) \delta(E - E_i(\mathbf{k}))$$

$$v_{i,q}(\mathbf{k}) = \frac{\partial E_i(\mathbf{k})}{\partial k_q}$$

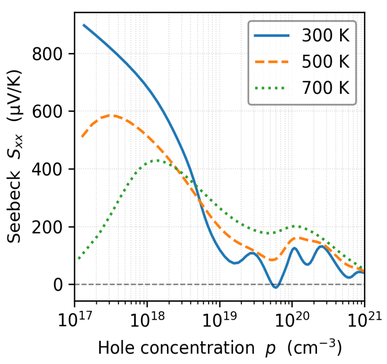
Organic Semiconductor Lattice Expansion

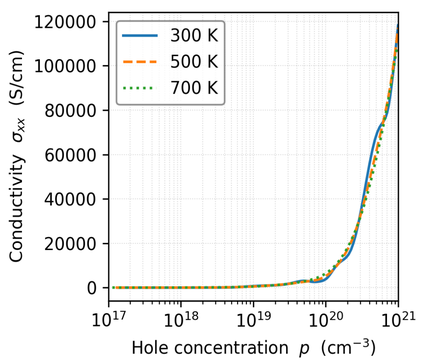




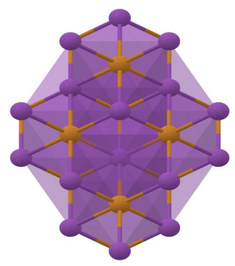


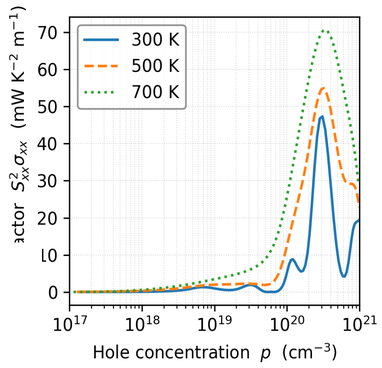
Thermoelectric Lattice Expansion

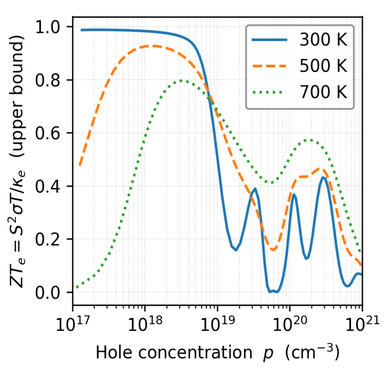


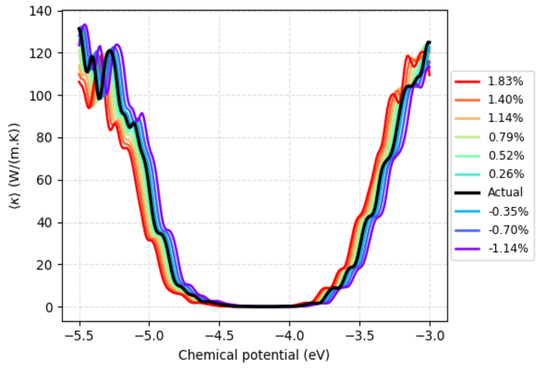


Bi2Te3

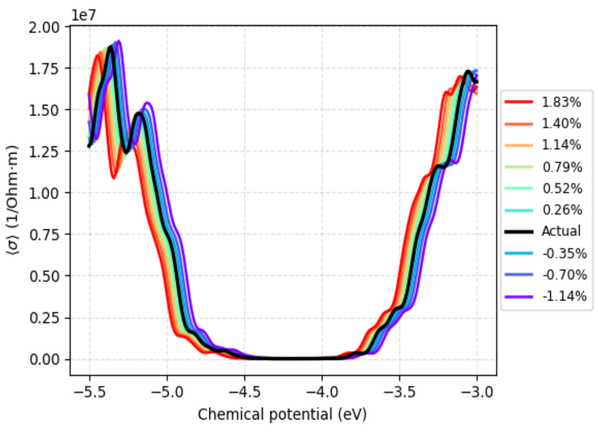




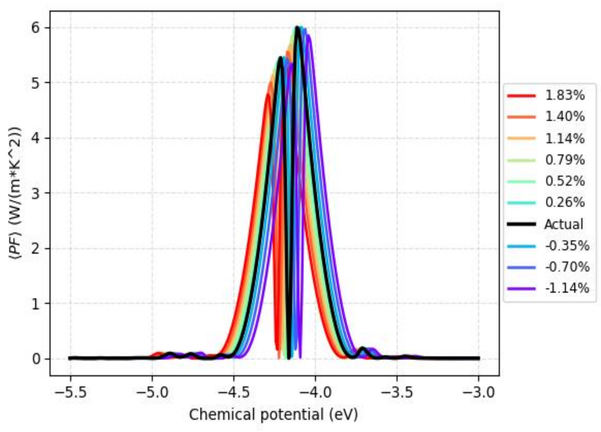




300 K KAPPA



300 K SIGMA



300 K PF

Discussion

- **Hole Conductivity** decreases with expansion
- **The figure of merit has** a dependency on the hole concentration
- **Electron-coupling** increases with the decrease in distance between the center of mass of the two monomers, but the repulsion force places them in equilibrium.
- **Kappa** decreases with expansion of hole carriers, but it increases with expansion of electron carriers.
- **Power factor** increases with temperature.

Conclusions

- Established computational workflow: **input file**, **HPC execution**, **output analysis** using **Bi_2Te_3.**
- Understanding **CRYSTAL23 input structure, keywords** and **Linux commands**
- Extending the study to doped **Bi_2Te_3** and **$C_{10}H_8$.**

References

Sosorev et al. (Mater. Des., 2020); Okamoto et al. (J. Am. Chem. Soc., 2020); Brédas et al. (PNAS, 2002); Coropceanu et al. (Chem. Rev., 2007); Giannini & Blumberger (Acc. Chem. Res., 2022); Zhu et al. (J. Phys. Mater., 2023); Podzorov et al. (Phys. Rev. Lett., 2004); Blancas et al. (npj Comput. Mater., 2024); Maul et al. (J. Phys. Chem. Lett., 2020); Erba et al. (J. Chem. Phys., 2014–2015); Finn et al. (Front. Electron. Mater., 2021); He & Tritt (Science, 2017); Erba et al. (J. Chem. Theory Comput., 2023); Dovesi et al. (CRYSTAL23 User's Manual, 2023).

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